QP CODE:25901822

MGU-BVOC (HONOURS) REGULAR EXAMINATION APRIL2026
SECOND SEMESTER
Multi disciplinary course-MG2MDCBAT100-PLANT PROPAGATION METHODS
(2025 Admission Onwards)

PART A

1. **Application of layering**
    - Propagation of woody perennial plants
    - Multiplication of plants with poor rooting ability
    - True to type propagation
    - Early bearing of crops
    - Propagation of climbing plants
2. Budding,grafting,cuttings,layering,micropropagation
3. **Detached Scion Grafting:** A method where the scion is completely separated from the mother plant before being joined to the rootstock. Examples include wedge grafting, tongue grafting, and cleft grafting.
4. **Parts of a Dicot Seed:**

1. **Seed Coat:** Outer protective layer (testa and tegmen).
2. **Embryo:** Consists of the radicle (future root), plumule (future shoot), and two cotyledons (food storage).

5. **Disease-Free Plant Production:** Propagation helps through **meristem culture** (which eliminates viruses because the growing tip is often pathogen-free) and by using certified disease-free mother plants or resistant rootstocks.
6. **Line Sowing (Drilling):** The process of sowing seeds in continuous lines at a controlled depth and uniform spacing, usually done using a seed drill or by hand in furrows.
7. **Compound Layering:** Also known as **serpentine layering**. A long branch is alternately covered with soil and left exposed at intervals. Roots form at every buried node, allowing multiple plants to be produced from a single shoot.

PART B

8. **Seed Preparation**

Extraction: Removing seeds from the fruit or pod.
Drying: Reducing moisture to ensure longevity.
Cleaning/Grading: Removing debris and selecting heavy, healthy seeds.
Seed Treatment: Using fungicides to prevent disease or techniques like scarification to break dormancy.

9. Importance of plant propagation in agriculture

- Ensures Continuous Supply of Planting Material
- Helps Maintain Genetic Purity and Uniformity

- Enables Improvement of Crop Varieties
- Conserves Elite and Rare Germplasm
- Facilitates Rapid Multiplication of Plants
- Essential for Perennial and Horticultural Crops
- Enhances Disease-Free Plant Production
- Supports Commercial Agriculture and Nurseries

10. DEVELOPMENT

Development includes all the changes in a plant from the germination of seed to senescence. Plants during their growth and development show different responses tothe environmental conditions and develop different structures. This ability is known as plasticity.
Example

- Heterophylly is seen in the cotton, coriander and buttercup plants.
- In cotton and coriander plants, the leaves in juvenile and adult plants are different morphologically

PART C

11. Seedling Care and Nursery Practices:

1. **Nursery Practices:** Includes site selection (well-drained, sunny), soil sterilization, and bed preparation (raised beds). Proper sowing depth and spacing are vital to prevent overcrowding.
2. **Seedling Care:**
    - **Watering:** Using a fine spray to avoid displacing seeds.
    - **Thinning:** Removing weak seedlings to reduce competition.
    - **Hardening off:** Gradually exposing seedlings to direct sunlight and reduced water before transplanting to prepare them for field conditions.
    - **Protection:** Using nets or fungicides to prevent "damping off" and pest attacks.

12. Plant Propagation and Methods:

The process of multiplying plant species to perpetuate them and maintain their population.
1. **Sexual Propagation (Seeds):** Involves the fusion of male and female gametes. It promotes genetic diversity but may result in offspring different from the parent.
2. **Asexual Propagation (Vegetative):** Uses vegetative parts like stems, roots, or leaves.
    - **Cuttings:** Rooting a piece of the parent plant.
    - **Grafting/Budding:** Joining parts from two different plants so they grow as one.
    - **Layering:** Inducing roots on a stem while it is still attached to the parent.
    - **Micropropagation:** Growing whole plants from cells in a sterile laboratory.